

Substrate SM Program — honest-state ledger v0.4 (Saturday late-morning batch absorption)

Keeper

2026-05-30 Saturday late-morning

Honest-state ledger v0.4 — Saturday late-morning batch absorption

Standing rule (Calibration #34 STANDING narrow framing): every cell marked DERIVED below is DERIVED modulo the one keystone bet (canonical-basis element of $U_q(B_2)$ IS a physical particle with that K-type). The headline-cap does not drop the conditionality.

v0.4 delta from v0.3

10 substantive late-morning landings absorbed; bulk-color gate (b) refined again; methodology stack steady at 19 layers.

Bulk-color gate (b) — first numerical count match

	v0.3 status	v0.4 status
(b) SU(3) QCD emergence	OPEN multi-week; route (C+D) COMMITTED v0.4	CANDIDATE COUNT MATCH via Lyra v0.5: explicit Toeplitz setup on $H^2(S)$ gives $8 = 3$ SO(3)-vector Toeplitz $T_a + 3$ pairwise $[T_a,$ $T_b] \in K(H^2(S))$ Hankel commutators + 2 K rank-2 Cartan. First numerical alignment for SU(3) 8-gluon emergence. Closure verification remains multi-week.
Conditions on (b)	C1 (Toeplitz-source) + C2 (AdS/CFT demotion)	C1+C2 carry forward + C3 (NEW): SO(3)-decomposition mismatch — $8 = 3+3+2$ is NOT su(3)'s canonical SO(3)-branching (which is $3+5$). v0.6 must (a) identify non-canonical SO(3) embedding, (b) verify non-canonical su(3) closure, OR (c) honestly mark as count-aligned-OPEN. C4 (NEW): $[T_a, T_b]$ linear-independence verification (Elie's lane).

Substantive advance from v0.4 (Lyra committed route C+D unified) to v0.5 (explicit Toeplitz setup + first count match). The count-match alignment is the strongest substrate-side advance on bulk-color since the (C+D) commitment.

L3 dynamics — Quasi-Eigentone Resolution B + Pair α channel structure

Claim	v0.4 status
Quasi-Eigentone framework (true/quasi/eigentone-in-vacuum)	FRAMEWORK carried from v0.1; SYNTHESIS-refined in v0.2
Spinor ³ channel structure (Grace 1.5 Pair α): {trivial, vector, adjoint} intermediate Casimirs {0, 4= rank^2 , 6= C_2 }	FRAMEWORK structural labels for the 3 lepton generations
Three bulk radial towers (Elie Toy 3627): $V_{(k,0)}$ vector $C=k(k+3)$; $V_{(k,k)}$ adjoint $C=2k(k+2)$; $V_{(k+1/2,k+1/2)}$ spinor $C=(k+1/2)(2k+5)$	RIGOROUS closed-form Casimirs
T190 exponent C_2 = gauge mediator (Resolution B)	SYNTHESIS-CANDIDATE with two-structures-burden flag
Two-structures-burden on Resolution B	OPEN: integer 6 = C_2 invoked in two distinct structural roles (intermediate-Casimir channel label + gauge-mediator adjoint Casimir). Parallels #414 v0.2 “one $h^\vee$ doing double duty” pattern. v0.3 must disambiguate: Reading 1 (over-determination, PASS) or Reading 2 (requires structurally-independent instances).
Quantitative T190 closed-form derivation $(24/\pi^2)^{C_2}$ from kernel integral on adjoint channel	OPEN multi-week (Elie post-affine-pin)

Substrate-arithmetic Mendeleev — Cal relation-reduction absorbed

Claim	v0.4 status
Substrate-arithmetic 8-18 cluster coincides with SM-gauge content (Grace Two-Route Scan v0.1) “12 = 4-route over-determined” ($n_C + g$; $\text{rank} + N_c + g$; $\text{rank} \cdot C_2$; $N_c \cdot \text{rank}^2$)	STRUCTURAL observation carried
Two-Route Scan v0.2 conditions	REDUCES TO 2-route under primary relations $C_2 =$ $N_c - \text{rank}$ and $n_C = \text{rank} + N_c$ per Cal #171. Net: $\text{rank} + N_c + g$ sum + $N_c \cdot \text{rank}^2$ product.
Candidate Casey-named-principle elevation	N1 (null-model toy) + N2 (cluster wording) + N3 (NEW Cal #171): route-count reformulation under structural-independence BLOCKED on N1 + N3

Same discipline as Strong-Uniqueness null-model recompute (single-residual framing, NOT multiplicative). Applied at substrate-arithmetic scale (Two-Route) and theorem-level scale (Strong-Uniqueness).

Affine pin — Langlands-duality reframe

Claim	v0.4 status
Affine pin: $B_2^{(1)}$ marks (1,1,2), $\delta = \alpha_0 + \alpha_1 + 2\alpha_2$, $h=4$	PINNED (convention) per Lyra v0.1 + Keeper refinement (v0.3 read: “same untwisted affine for $n=2$ since $B_2 \cong C_2$ ”)

Claim	v0.4 status
$B_2^{(1)} \leftrightarrow C_2^{(1)}$ IS Langlands duality at affine level (Lyra B3 session)	STRUCTURAL REFRAME: not just “same untwisted affine via finite-isomorphism” but a deeper Langlands-dual structural reading. The two pins describe the substrate from dual sides. Geometric-Langlands axis active.

L1 engine — Drinfeld/CPT + bulk-color algebraic pre-reserved

Claim	v0.4 status
Engine v0.2 PASS	CARRIED (Lyra L1-L4 + Elie E0-E10)
Engine v0.3 §6 candidate (Drinfeld double + CPT, Toy 3617)	STRUCTURAL pre-reserved; K1·2 anchor reserved
Engine v0.3 §7 candidate ($SO(5) \supset SO(3) \times SO(2)$ bulk-color algebraic, Toy 3620)	STRUCTURAL pre-reserved; K1·4 anchor reserved; cross-confirms bulk-color (D) $SO(3)$ sub-vector
Engine v0.3 consolidation doc filing (Elie)	PENDING — the actual v0.3 doc, not the audit-support doc

B-track mathematical excavation (Lyra session, 6 axes touched)

Axis	v0.4 status
B1 quantum groups at $q=2$ deeper	FRAMEWORK — 6 layers mapped (Frobenius, Steenrod, mod-2 cohomology, modular reps, characteristic-2 algebra)
B2 Macdonald-Koornwinder corners	FRAMEWORK — 2 corners occupied + others mapped
B3 Geometric Langlands	FRAMEWORK — $B_2^{(1)} \leftrightarrow C_2^{(1)}$ Langlands duality at affine level (reframes affine pin)
B4 MTC from R-matrix	FRAMEWORK — all MTC ingredients present; S-matrix non-degeneracy OPEN
B5 VOA / Monster	STRUCTURAL — 24/26 align with Moonshine integers; K57 connection
B8 cross-domain bounded-symmetric-domain uniqueness	FRAMEWORK — D_{IV}^5 uniquely selected per-domain (Lyra check)
B6 K-theory of D_{IV}^5	PENDING
B7 higher Wallach baryon spectrum	PENDING

A3 L5 absolute scale — honest negative on 4 candidates

Claim	v0.4 status
L5 absolute scale + Higgs/EWSB	OPEN; HONEST NEGATIVE on 4 candidates (Lyra session) — genuinely open; least-developed L-axis remains genuinely undeveloped

This is the right kind of progress: Lyra ruled out four explicit candidates and admitted L5 is genuinely open. Honest negatives are framework-strengthening (per cumulative methodology pattern).

Saturday batch carry-forwards (from v0.3, still standing)

- Three independent quantitative falsifiers LIVE: F1 PMNS 3/3 within 1σ (JUNO/DUNE 2025-2030); T1 lepton count $24 = 4 \times 2 \times 3$ MATCHED; Bell sub-Tsirelson $1/2^N c = 1/8$ (T2469 SCMP).
- Strong-Uniqueness v1.1: 13 RIGOROUSLY CLOSED + META T2467 + C18 (Born/Gleason measure-FORCED theorem) + C19 (ρ -vector pinning); Keeper null-model recompute (single-residual framing); v1.2 pending Lyra absorption.
- Calibration #34 STANDING narrow framing (headline-cap-conditionality); discipline stack 5 elements (#27 + #29 + #32 + #33 + #34); methodology stack 19 layers.
- Periodic Table v0.7 absorbed: 6-tuple per-cell taxonomy with stability class (true/quasi/eigentone-in-vacuum); mass-eigenstate-vs-flavor ν framing.
- K-types-are-particles falsifier triad (Cal + Lyra + Grace): three layers PASS; T1 evolved from Cal addendum (target=24 a-priori pinned) into Lyra T1 v0.1 (pure-QG dim 4 + dictionary-combined 24 both MATCHED).
- 6-domain $225 = (N_c \cdot n_C)^2$ fingerprint (Bergman + Ag Debye + Cu/Ag + Ag/Pb + cosmological c_{reg} + CKM $|V_{cb}|$).

Open gates carrying forward (consolidated)

Gate	Status
Bulk-color gate (b)	CANDIDATE COUNT MATCH ($8 = 3+3+2$) via Lyra v0.5; C3 SO(3)-decomposition concern + C4 $[T_a, T_b]$ independence are v0.6 conditions; closure verification multi-week
Bulk-color gate (a) counting layer	ADDRESSED via SO(3) sub-vector; cross-confirmed by engine v0.3 §7 candidate
Generation-mechanism identification	TENSION RELIEVED, NOT CLOSED; count forced ($h^\nu = 3$); two-structures via SO(3)-geometric color + algebraic spinor ³ channels
Per-particle quark/gauge dictionary (L2/L6)	ASSIGNED; flips when bulk-color closes per #418
L4 v0.2 per-particle mass	UNBUILT; Resolution B gives concrete target $(24/\pi^2)^{C_2}$; multi-week kernel-integral
L5 absolute scale + Higgs/EWSB	OPEN; 4 candidates honest-negative'd Saturday (least-developed L-axis)
Engine v0.3 consolidation doc	PENDING Elie filing
A1 PRIMARY paper Cal quick-verify + proof-stage genus citation	PENDING
Strong-Uniqueness v1.2	PENDING Lyra absorbing Keeper null-model recompute + Cal cold-read
Quasi-Eigentone v0.3	PENDING Lyra disambiguating Resolution B reading (1 over-determination vs 2 structurally-independent $C_2=6$ instances)
C12-C14 T-number/META audit	PENDING (Cal #156 + Cal #99 framing)
Two-Route Scan v0.2	PENDING Grace N1 (null-model toy) + N2 (cluster wording) + N3 (relation-reduction per Cal #171)
Casey-named principle candidates from Two-Route SP-30 outreach drafts (eigentone, Cs-137, W-32)	BLOCKED on N1 + N3 PENDING Casey send-signals (#330)

Saturday batch summary (cumulative)

- Tier-gates filed Saturday: 16 substantive items dispositioned across 8 consolidated docs (afternoon batch 7 items + late-morning batch 4 items + falsifier triad 3 items + Cal #34 ratification + Two-Route correction note)
- Calibration #34 STANDING under narrow framing

- Honest-State Ledger v0.4 with full Saturday absorption
- Bulk-color frontier: v0.2 (CANDIDATE $SO(3)$ sub-vector) $\rightarrow$ v0.4 (route C+D COMMITTED) $\rightarrow$ v0.5 (first numerical count match $8 = 3+3+2$) — substantive advance through the day
- Multi-week work continues: bulk-color $SU(3)$ closure verification + L4 v0.2 mass spectrum (Resolution B kernel-integral target) + L5 absolute scale (genuinely open) + B-track mathematical excavation (6 axes active)
- Cross-CI discipline standing: Cal #27 firing at peak-convergence consistently caught (relation-reduction brake on Two-Route 4-route claim); Cal #34 conditional-tag-with-headline ratified; null-model framing corrected at both arithmetic and theorem scales

— Keeper. Ledger v0.4; bulk-color v0.5 first numerical count match is the headline; Resolution B two-structures-burden + $SO(3)$ -decomposition C3 concern carry forward as v0.6/v0.3 multi-week burdens; Cal relation-reduction brake applied at Two-Route scale; methodology infrastructure mature.